

Technical Skills

Languages: Java, Python, C/C++, JavaScript, SQL, ARM Assembly, Bash

Architecture & Engineering: Clean Architecture, Domain-Driven Design (DDD), OOP, UML, Design Patterns, Agile

Data Science & AI: PyTorch, scikit-learn, Pandas, NumPy, SciPy, Dask, Big Data Analytics

Web, Backend & Databases: REST APIs, Flask, React, HTML/CSS, Tailwind, MySQL

Systems & Security: TCP/IP, Multithreading, Wireshark, AppSec Basics, Authentication

Tools & Cloud: Git/GitHub/GitLab, Docker, Linux/Unix, CMake, JUnit, PyTest, Cloud Basics

Internship

Federal Government of Canada | Environment and Climate Change

May 2026 – Aug 2026

Data Science Intern & Anomaly Detection

Quebec City, QC

- Engineered an automated unsupervised machine learning pipeline (Isolation Forest, LOF, DBSCAN) to detect spatial anomalies and “bullseye” artifacts in precipitation maps, replacing a 30-day delayed manual visual inspection with a rapid early-detection system.
- Applied advanced statistical modeling, probability distributions, and spatial autocorrelation metrics for rigorous Exploratory Data Analysis (EDA), data preprocessing, and feature engineering on massive-scale meteorological datasets.
- Orchestrated high-performance distributed computing and geospatial analysis using **Python, scikit-learn, SciPy, pandas, NumPy, xarray, GeoPandas, and Dask**, while overhauling the entire project documentation.

Involvement

Research Project – SGD Beyond

2025 – 2026

Academic Machine Learning Club

- Researched stochastic optimization, reviewing modern SGD extensions and non-convex optimization landscapes.
- Contributed to a theoretical SGD enhancement integrating controlled noise, implemented and validated in Python (PyTorch).

CANLock Project – Vehicle Cybersecurity

2025

Academic & Industrial Collaboration (Thales Partnership)

- Analyzed vehicle network data and explored AI approaches for anomaly and attack detection on CAN buses.
- Collaborated in a multidisciplinary team across cybersecurity, data science, and embedded systems to secure connected vehicles.

Engineering Projects

BâtiBloc | Java, Swing, JavaFX, PDFBox

2026

Winner of the Yves-Roy Award (Best Departmental Object-Oriented Software Engineering Project)

Nominated – Best Software Engineering Project of the Year (2025/2026)

- Engineered a full-stack construction tool bridging architectural PDF blueprints and job-site reality: import via PDFBox with drag & drop, imperial scale calibration, and multi-shape zone definition (rectangular, triangular, truncated) including openings and roofs.
- Built a complete block placement simulation engine handling sub-6in cuts, load transfers, and lintels, paired with a cost estimation dashboard (per-zone/view/global) with editable unit prices and exportable PDF reports.
- Delivered hybrid 2D/3D visualization (Swing + JavaFX + FlatLaf) with multi-material OBJ rendering, full 3D manipulation, persistent `.batibloc` format, filterable history, Undo/Redo (Command pattern), performance mode, and an integrated chatbot.
- Architected a clean layered system (Domain / Application DTO / Presentation) ensuring full 2D/3D interoperability and maintainable separation of concerns.

Medical Case Simulator | Python, Flask, MySQL, Vanilla JS

2026

- Architected a 3-tier web application for medical clinical case simulations, featuring a responsive, WCAG AA-compliant frontend, a Python/Flask REST API, and a centralized relational database.
- Designed and optimized a highly normalized (BCNF) MySQL database schema utilizing complex SQL queries, triggers, stored procedures, and strategic indexing for real-time community forum analytics.
- Enforced strict application security mechanisms including bcrypt password hashing, parameterized SQL queries to prevent injections, secure session management, and asynchronous AJAX client-side validations.

Secure Web Automation (ConUHack26) | Python, Selenium

2026

- Developed an autonomous agent successfully bypassing anti-bot and behavioral tracking mechanisms.
- Reverse-engineered client-server exchanges via Wireshark and enforced robust application security.

Banking System | C++, Qt

2025

- Built an OOP banking application enforcing secure transaction processing with robust hierarchical exception handling.

Education

Université Laval

Bachelor of Software Engineering, 3rd Year | GPA: 3.5 / 4.3

Quebec City, QC

Brock University

Student Exchange Program

St. Catharines, ON